Homework #8: Operating Systems (15 points)

Due to Gradescope by 11:45 PM on Tuesday, October 11th

This assignment cannot be submitted late

You need to submit a pdf to Gradescope; failure to assign questions to pages will result in a 10% deduction on your grade

Homework Goal: Review and practice the material we have covered for the Operating Systems unit.

Vocab and Definitions

1. Practice some vocab from this chapter! (1 point)
   GUI stands for graphical user interface.
   The ready queue holds the line of processes that are ready to run.

2. List the four major tasks that the operating system handles that we discussed in class. (1 point)
   1. Provide user interface
   2. Offer System Security & Protection
   3. Manage Resources Efficiently
   4. Use Resources Safely

3. Define superusers and give an example of why we need them, besides for maintaining password files. (1 point)
   Superuser - a user who has more privlages then another

   We need them for business or schools so a regular user can't install software that could harm the computer or network.

4. What is the difference between a program and a process? (1 point)
   a program contains a set of intructions and does not use any rescoures only resides in memory

   a process is a program in executing using resources efficiently

5. Suppose I have a process in the BLOCKED state, waiting to get a file that your process has. You have a process in the BLOCKED state, waiting to get a file that my process has. What is this situation called, and how can it be resolved? (1 point)

Deadlock

resolution: one process uses resources at a time and the other waits for its turn

Process States

6. The following table provides a grid of the possible process states. Name each valid transition, coming FROM the left column and going TO the right. If a transition does not occur, *leave the box blank*. Think about all possible transitions. For example, when does dispatch occur? What state do we come from? To which state do we go? Write "dispatch" in that box. It may help to draw out the diagram below the table. (3 points)

OS State Transition Table		**To**			
		Ready	**Running**	**Blocked**	**Exit**
From	**New**	admit			
	Ready		Dispatch		killed
	Running	Timeslice		Event Requested	Finish or Error
	Blocked	Event occurs			killed

7. Suppose you have two processes to run: a CPU-intensive process and an I/O-intensive process. If you are trying to run these processes on a heavily loaded single processor machine (it has a lot of other processes that also need to run), then what state will each process spend most of its time in (besides RUNNING)? Justify your response. (1 point)

   (a) CPU-intensive process

A CPU-intensive process spends most time in the ready state because it needs to be in running but gets booted out after the timeslice

   (b) I/O-intensive process

A I/O-intensive process spends time in blocked because it is waiting for an event to occur.

Processor Performance

8. Anastasia notices her system response time is very slow. To improve the performance, she decides to reduce the system's time slice from 100 ms to 10 ms. Her system takes 5 ms to change from one process to another (counted as part of that process's time slice). Is Anastasia's time slice reduction a good way to improve performance? Justify your response. (1 point)

No, because to close a process and start a new one taks 5ms that means each process only gets 5ms of running time.

9. Processor utilization is the percentage of time that the processor spends doing *useful* work (such as executing arithmetic instructions). Assume that an individual process in a system spends about 34% of its time waiting for I/O operations to be completed (*not* useful work). When one process is waiting on I/O, the processor can execute another process that is not waiting for I/O. For both parts (a) and (b), show your work. (2 points)

(a) Assume that every process spends about 54% of its time waiting for I/O operations to be completed (not useful work). How many processes should we keep in memory if we want processor utilization to be at least 97%?

$1 - 0.54^x > 0.97$

$0.03 > 0.54^x$

$x \geq 6$ programs

(b) Processes A, B, and C spend 37%, 45%, 27% of their time, respectively, waiting for I/O operations to be completed. What is the processor utilization if Process A, B, and C are loaded into memory?

$1 - (.37)(.45)(.27)$

$= 95.5\%$

Virtual Memory

10. Answer the following questions using this virtual memory table. (2 points)

Physical Memory Address	Virtual Memory Address	Allocated To:
0 - 15	0 - 15	Program A
16 - 31	0 - 15	Program B
32 - 47	16 - 31	Program A
48 - 63	32 - 47	Program A
64 - 79	?	Program B

(a) What range of addresses should replace the question mark in the above table?

16 - 31

(b) Program A wants to access a variable stored in virtual address 20. What physical address is this variable stored in?

$PA = (20 - 16) + 32$

$= 36$

0	0
15	15
32	16
33	17
34	18
35	19
36	20

11. Consider a Python program that is stored in your computer's memory. For this program, what is the first virtual address (give a number)? Is this guaranteed to be the same as the first physical address? Why or why not? (1 point)

First Virtual Address: 0

No, it is not guarenteed to be at the first physical address because their could be programs before it like in the example above where program B is not the same as the physical location.